ELECTRO-OPTICAL SYSTEM PERFORMANCE

Sensitivity Analysis Improvement Plan

Practical engineering recommendations for your TRM-based application

01 **UNCERTAINTY**

Correct interval language and calculation assumptions

02 **PARAMETER SWEEP**

Replace stale plots with DRI-versus-parameter curves

03 **LIMITING FACTORS**

Separate performance constraints from uncertainty drivers

BASED ON

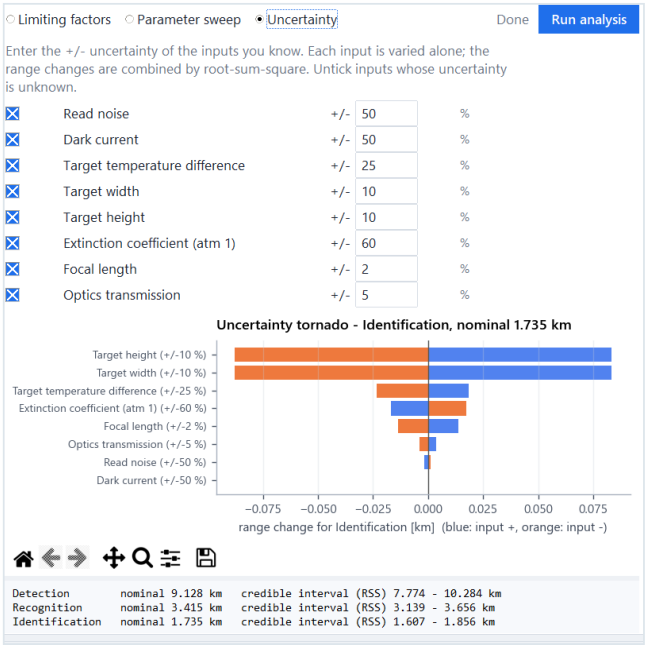
Two application screenshots and TRM4.v3 methodology

Scope: existing Johnson/DRI model only; no adaptive criteria

Technical design note | 09 October 2026

Current state: what the screenshots show

Observed interface behavior is distinguished from recommendations that require code-level verification.



A. Uncertainty tab: RSS tornado



B. Sweep tab: previous tornado remains visible

Observation	Engineering assessment
What is already good	Three separate modes; selective input uncertainties; a recognizable tornado visualization; baseline DRI results.
Critical terminology	The label "credible interval (RSS)" should not be used unless a probabilistic interpretation has been justified.
Results-state clarity	In the Sweep screenshot, the displayed plot is still an Uncertainty tornado. A pending/previous-result state must be explicit.
Ranking interpretation	Target width and height dominate the displayed Identification tornado; read noise and dark current have very small effects. This is not automatically a model defect.

> **SOURCE STATUS**
>
> The screenshots show user-entered uncertainty percentages and reported DRI ranges. They do **not** reveal whether the code fully recalculates the TRM engine, how input uncertainty is defined, or how the RSS result is computed. Those behaviors must be verified.

1 | Uncertainty: fix the mathematics first

Recommended minimum change: separate deterministic variation from statistically valid uncertainty propagation.

1.1 Three different outputs - do not mix them

Method	Inputs required	Correct output label
One-at-a-time variations	Baseline and explicit -/+ perturbation bounds	Range change from each input (-/+), not a probability interval
RSS propagation	Standard uncertainties, local sensitivities, independence (or covariances)	Approximate combined standard uncertainty (or expanded uncertainty if factor is defined)
Monte Carlo	Probability distributions, dependencies, sample settings	Percentiles or estimated probability interval with stated coverage

1.2 Correct uncertainty calculation

For independent inputs with defined standard uncertainties, estimate the combined standard uncertainty of a DRI range **R** as:

$$u(R) \approx \sqrt{\sum_i \left[\left(\frac{dR}{dx_i}\right)^2 \cdot u(x_i)^2 \right]}$$

When inputs are correlated, covariance terms must also be included. A user-entered "+/- 50%" is **not** automatically a one-standard-deviation uncertainty; the interface must ask what it represents.

1.3 Required UI changes

- **Uncertainty meaning:** bound, 1-sigma, 95% interval, or user-defined distribution.
- **Per-input distribution:** Normal, Uniform, Triangular, or Lognormal (where physically appropriate).
- **Correlation / coupling:** support defined correlations or linked parameter groups.
- **Method selection:** RSS approximation versus Monte Carlo; show assumptions and run settings.
- **Correct heading:** replace "Credible interval (RSS)" with an accurate method-dependent term.

> **MODEL SAFEGUARD**
>
> Nonlinear DRI relationships can make plus and minus effects asymmetric. Very large perturbations, such as +/- 60% atmospheric extinction, justify checking the full recalculated response and using Monte Carlo when probabilistic assumptions are available.

2 | Redesign the uncertainty results

Keep the Tornado chart. Improve what it communicates and make every value traceable.

2.1 The results should answer three questions

- **Which input changes DRI most?** Rank by absolute change in range over the stated perturbation.
- **Which input contributes most to result uncertainty?** Rank by modeled uncertainty contribution.
- **How sensitive is the system itself?** Rank by normalized local sensitivity per 1% input change.

2.2 Detailed Tornado behavior

Feature	Recommended behavior
Axis	Centered on zero for delta-range view; provide an optional absolute-range view in km.
Tooltip / selection	Show baseline input, low/high input, recomputed D/R/I, delta km and delta %.
Color legend	Use explicit labels "input decreased" and "input increased"; color must not imply improvement by itself.
Ordering	Sort by the maximum absolute range effect, with a setting to sort by normalized sensitivity.
Outcome selector	Detection / Recognition / Identification without losing parameter settings.
Missing effects	Show "below display resolution" rather than silently displaying an apparently absent bar.

2.3 Improve the results summary

FROM YOUR SCREENSHOT (reported values, not independently validated)

Detection	Recognition	Identification
9.128 km	3.415 km	1.735 km
7.774 - 10.284 km	3.139 - 3.656 km	1.607 - 1.856 km

DISPLAY RULE

Put nominal DRI results and method-labeled intervals in distinct, readable result cards. Display source assumptions, selected inputs, and method details under an expandable "Calculation details" section.

3 | Parameter Sweep: show the actual sweep

The current tab supports parameter, start, end, and point count. The results area should reflect that mode immediately.

3.1 Suggested workflow

1. **Select parameter**
   Example shown: horizontal detector pitch [um].
2. **Set range and sampling**
   5 to 15 um, 3 points -> 5, 10, and 15 um.
3. **Select fixed-parameter policy**
   Decide what is held fixed; validate physical consistency.
4. **Run full calculations**
   Recompute relevant model components for each configuration.
5. **Plot D/R/I against parameter**
   Display curves and a numerical table; mark baseline.

3.2 Plot and table behavior

Current behavior	Improved behavior
Sweep tab can still display prior tornado	Switch graph to "DRI range vs parameter"; if inputs changed, show "Results not updated - Run analysis".
Only start/end/points shown	Add linear/log sampling, unit display, baseline marker, current engine mode and hold-constant assumptions.
No per-run output table visible	List parameter value, DRI ranges, percentage delta, and any invalid-run diagnostic.
Single parameter only	Keep 1D as default; introduce 2D grid / heatmap later as an advanced feature.

ENGINEERING WARNING

Changing detector pitch while keeping detector count and focal length fixed changes the calculated FOV/IFOV. A physically comparable detector swap may also change sensitivity, fill factor and noise; those changes must not be silently inferred from pitch alone.

4 | Protect physical consistency

Dependency-aware re-evaluation prevents misleading performance changes when two parameters are not independent.

4.1 Parameter dependency rules

Parameter varied	What must be reconsidered
Detector pixel pitch	IFOV, Nyquist, detector active area, detector sampling MTF, FOV (if focal length/resolution remain fixed), and possible detector noise assumptions.
Focal length	IFOV/FOV, sampling projection, pupil and F-number relationship, and applicable optical MTF profile.
F-number / clear aperture	Collected radiation and diffraction MTF. If using measured camera NETD, ensure it is valid for the optical configuration.
Optical MTF curve	Replace the corresponding optical term. Do not multiply diffraction twice if already included in the imported MTF.
Atmospheric extinction	Recalculate spectral/effective transmission, range-dependent signal and task limits.
Target geometry or signature	Recompute projected target dimensions, task spatial frequencies and range-dependent signal contrast.
Read noise / dark current	Recompute applicable signal-noise chain; warn if sensor mode uses a measured aggregate NETD instead.

4.2 A minimal simulation-engine contract

```
base_case = validate_configuration(current_camera, target, atmosphere)
base_result = calculate_full_performance(base_case)
for value in parameter_grid:
    case = clone(base_case); case[parameter] = value
    case = apply_dependency_policy_and_validate(case)
    result = calculate_full_performance(case)
    save(value, result.D, result.R, result.I, result.diagnostics)
```

NOT A SIMULATED IMAGE

These analyses are numerical recalculations of TRM-based performance. Generating realistic output images requires an additional image-formation simulation with scene radiance, PSF, sampling, noise and processing. Keep the two tools separate.

5 | Limiting Factors and advanced studies

Provide decisions, not just raw sensitivities. Keep optional complex analyses out of the default view.

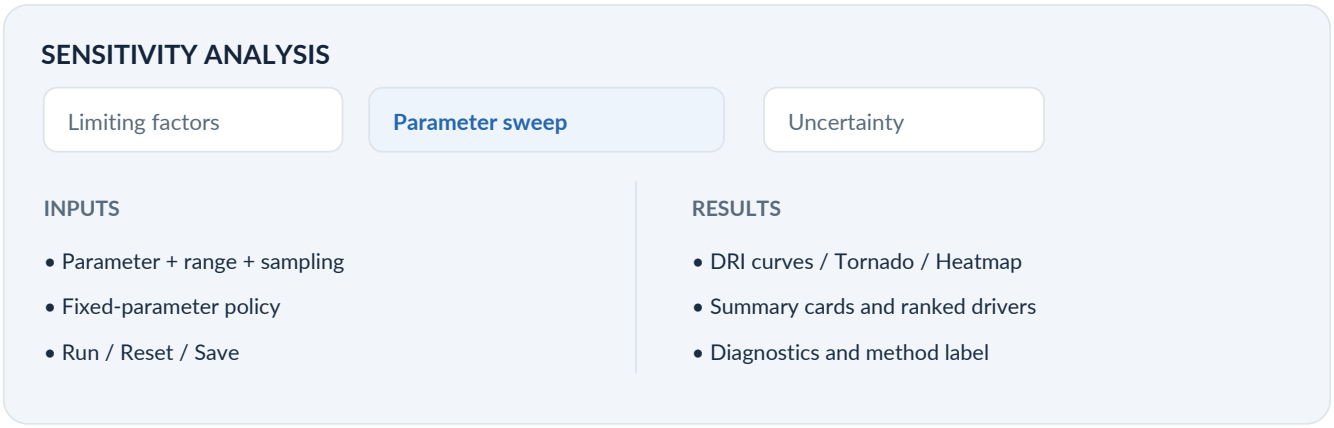
5.1 Distinguish three engineering questions

Question	Output	Use
What limits nominal range?	Optical resolution / sampling / noise / atmosphere / target contrast diagnostics	Performance bottleneck review
Which input is most uncertain?	Uncertainty contribution ranking and distribution impact	Confidence in predicted results
Which design variable is worth improving?	Local normalized sensitivity and feasible design variation	Design optimization and trade studies

5.2 Recommended optional features

- **Two-parameter sweep:** a heatmap for two physically compatible variables, for example target contrast versus extinction; mark invalid combinations.
- **Scenario comparison:** save multiple baseline configurations; overlay DRI curves and summarize deltas.
- **Report export:** PDF/CSV with original inputs, variations, algorithm options, results, diagnostics and date.
- **Run history:** retain parameter values and mark a result as stale when any source input changes.

5.3 Proposed compact screen hierarchy



UX PRINCIPLE

Use one consistent top-level layout for all three modes, but change the central plot and controls to match the selected analysis. Keep advanced numerical controls behind an expandable panel.

6 | Implementation priorities and validation

A developer-ready implementation plan. Items are recommendations based on the visible UI and referenced TRM theory.

6.1 Prioritized change backlog

Priority	Change	Acceptance criterion
P0	Rename RSS output; define uncertainty semantics	No "credible interval" for raw RSS; method and input interpretation visible.
P0	Enforce plot/result state by mode	Sweep never shows an unlabeled uncertainty plot as if it were new sweep output.
P0	Full-model recomputation plus dependency validation	Every modified configuration is checked and appropriate affected modules are recalculated.
P1	Add sweep DRI curves, data table and baseline marker	Numeric table and chart agree at every sampled point.
P1	Interactive tornado and normalized rankings	Each bar reveals baseline, +/- runs and resulting DRI deltas.
P1	Readable DRI cards and report export	Method, units, assumptions and run metadata appear in outputs.
P2	Monte Carlo + correlations	Known distributions/correlations yield reproducible percentile output.
P2	Two-parameter heatmap + comparative scenarios	Invalid combinations are rejected; full recalculation per cell.

6.2 Verification checklist

- **Zero perturbation:** the result exactly reproduces the nominal DRI values.
- **Sign and direction:** attenuation increasing should not accidentally be plotted as "improvement".
- **Symmetric perturbation test:** linear toy models agree with RSS theory; nonlinear cases show asymmetry.
- **Numerical convergence:** sweep resolution and root-finding tolerance do not cause unstable DRI ranges.
- **State and reproducibility:** saved inputs and fixed random seeds reproduce the same Monte Carlo summary.
- **System integrity:** externally imported MTF and measured NETD are not double counted.

6.3 Traceable technical basis

User-provided: Uncertainty and Parameter Sweep screenshots dated 09 October 2026 (embedded on page 2).
Fraunhofer IOSB: TRM4.v3, *Parameter Description and Model Documentation*, Technical Report 2021/04; especially Sections 3.11-3.12 (perceived thresholds/range), 4.1-4.3 (MTF/AMOP).
Additional statistical guidance: JCGM 100:2008 (GUM) and JCGM 101:2008 (Monte Carlo propagation). These references explain uncertainty methods; they are not claims that the current application implements them.